

Engineering Design Optimization Problems

I. Welded Beam Design Problem

The Welded Beam Design (WBD) problem is a minimization problem aiming to reduce the manufacturing cost of a welded beam. The problem involves four design variables: the weld thickness ($l_1 = h$), the length of the welded joint ($l_2 = l$), the width of the beam ($l_3 = t$), and the thickness of the beam ($l_4 = b$). The design is subject to constraints on shear stress (τ), bending stress (σ), buckling load (P_c), and end deflection (δ).

Variables:

$$\vec{l} = [l_1, l_2, l_3, l_4] = [h, l, t, b] = [x_1, x_2, x_3, x_4]$$

Objective Function:

$$\min f(\vec{l}) = 1.10471l_1^2l_2 + 0.04811l_3l_4(14.0 + l_2)$$

Constraints:

$$\begin{aligned} s_1(\vec{l}) &= \tau(\vec{l}) - \tau_{\max} \leq 0 \\ s_2(\vec{l}) &= \sigma(\vec{l}) - \sigma_{\max} \leq 0 \\ s_3(\vec{l}) &= \delta(\vec{l}) - \delta_{\max} \leq 0 \\ s_4(\vec{l}) &= l_1 - l_4 \leq 0 \\ s_5(\vec{l}) &= P - P_c(\vec{l}) \leq 0 \\ s_6(\vec{l}) &= 0.125 - l_1 \leq 0 \\ s_7(\vec{l}) &= 1.10471l_1^2 + 0.04811l_3l_4(14.0 + l_2) - 5.0 \leq 0 \end{aligned}$$

Where:

$$\begin{aligned} \tau(\vec{l}) &= \sqrt{(\tau')^2 + 2\tau'\tau''(l_2/R) + (\tau'')^2} \\ \tau' &= \frac{P}{\sqrt{2}l_1l_2}, \quad \tau'' = \frac{MR}{J}, \quad M = P(L + l_2/2) \\ R &= \sqrt{\frac{l_2^2 + (l_1 + l_3)^2}{4}} \\ J &= 2 \left\{ \sqrt{2}l_1l_2 \left[\frac{l_2^2}{12} + \frac{(l_1 + l_3)^2}{4} \right] \right\} \\ P_c(\vec{l}) &= \frac{4.013El_3l_4^3}{6L^2} \left(1 - \frac{l_3\sqrt{E}}{8LG} \right) \end{aligned}$$

Parameters: $\sigma_{\max} = 30000$ psi, $P = 6000$ lb, $L = 14$ in, $\delta_{\max} = 0.25$ in, $E = 30 \times 10^6$ psi, $\tau_{\max} = 13600$ psi, $G = 12 \times 10^6$ psi.

Variable Ranges:

$$0.1 \leq l_1, l_4 \leq 2, \quad 0.1 \leq l_2, l_3 \leq 10$$

II. Compression Spring Design Problem

The goal of the Compression Spring Design (CSD) is to minimize its weight $f(x)$ under four inequality constraints including minimum deflection, shear stress, surge frequency, and limit on outer diameter. It has 3 design variables: wire diameter d (x_1), mean coil diameter D (x_2), and the number of active coils N (x_3).

Objective Function:

$$\min f(x) = (N + 2)Dd^2$$

Constraints:

$$\begin{aligned} g_1(x) &= 1 - \frac{D^3 N}{71785d^4} \leq 0 \\ g_2(x) &= \frac{4D^2 - dD}{12566(Dd^3 - d^4)} + \frac{1}{5108d^2} - 1 \leq 0 \\ g_3(x) &= 1 - \frac{140.45d}{D^2 N} \leq 0 \\ g_4(x) &= \frac{D + d}{1.5} - 1 \leq 0 \end{aligned}$$

Variable Ranges:

$$0.05 \leq x_1 \leq 2, \quad 0.25 \leq x_2 \leq 1.3, \quad 2 \leq x_3 \leq 15$$

III. Pressure Vessel Design Problem

The objective of Pressure Vessel Design (PVD) is to minimize the total cost $f(x)$ (including material, forming, and welding) while meeting production needs. The 4 design variables are: thickness of the shell T_s (x_1), thickness of the head T_h (x_2), inner radius R (x_3), and length of the vessel without heads L (x_4). T_s and T_h are integer multiples of 0.625, while R and L are continuous variables.

Objective Function:

$$\min f(x) = 0.6224x_1x_3x_4 + 1.7781x_2x_3^2 + 3.1661x_1^2x_4 + 19.84x_1^2x_3$$

Constraints:

$$\begin{aligned} g_1(x) &= -x_1 + 0.0193x_3 \leq 0 \\ g_2(x) &= -x_2 + 0.00954x_3 \leq 0 \\ g_3(x) &= -\pi x_3^2x_4 - \frac{4}{3}\pi x_3^3 + 1296000 \leq 0 \\ g_4(x) &= x_4 - 240 \leq 0 \end{aligned}$$

Variable Ranges:

$$0 \leq x_1, x_2 \leq 99, \quad 10 \leq x_3, x_4 \leq 200$$

V. Speed Reducer Design Problem

In mechanical systems, a speed reducer is a crucial part of a gearbox used in various applications. The objective is to minimize the weight of the speed reducer subject to 11 constraints. The 7 variables are: face width b (x_1), module of teeth m (x_2), number of teeth on pinion z (x_3), length of the first shaft between bearings l_1 (x_4), length of the second shaft between bearings l_2 (x_5), diameter of the first shaft d_1 (x_6), and diameter of the second shaft d_2 (x_7).

Objective Function:

$$f(X) = 0.7854x_1x_2^2(3.3333x_3^2 + 14.9334x_3 - 43.0934) \\ - 1.508x_1(x_6^2 + x_7^2) + 7.4777(x_6^3 + x_7^3) + 0.7854(x_4x_6^2 + x_5x_7^2)$$

Constraints:

$$g_1(X) = \frac{27}{x_1x_2^2x_3} - 1 \leq 0 \\ g_2(X) = \frac{397.5}{x_1x_2^2x_3^2} - 1 \leq 0 \\ g_3(X) = \frac{1.93x_4^3}{x_2x_6^4x_3} - 1 \leq 0 \\ g_4(X) = \frac{1.93x_5^3}{x_2x_7^4x_3} - 1 \leq 0 \\ g_5(X) = \frac{\sqrt{(745x_4/(x_2x_3))^2 + 16.9 \times 10^6}}{110x_6^3} - 1 \leq 0 \\ g_6(X) = \frac{\sqrt{(745x_5/(x_2x_3))^2 + 157.5 \times 10^6}}{85x_7^3} - 1 \leq 0 \\ g_7(X) = \frac{x_2x_3}{40} - 1 \leq 0 \\ g_8(X) = \frac{5x_2}{x_1} - 1 \leq 0 \\ g_9(X) = \frac{x_1}{12x_2} - 1 \leq 0 \\ g_{10}(X) = \frac{1.5x_6 + 1.9}{x_4} - 1 \leq 0 \\ g_{11}(X) = \frac{1.1x_7 + 1.9}{x_5} - 1 \leq 0$$

Variable Ranges:

$$2.6 \leq x_1 \leq 3.6, \quad 0.7 \leq x_2 \leq 0.8, \quad 17 \leq x_3 \leq 28, \quad 7.3 \leq x_4 \leq 8.3 \\ 7.8 \leq x_5 \leq 8.3, \quad 2.9 \leq x_6 \leq 3.9, \quad 5.0 \leq x_7 \leq 5.5$$

VI. Tubular Column Design Problem

The goal is to design a uniform tubular column to support a compressive load $P = 2500$ kgf at minimum cost. The mean diameter d (x_1) and thickness t (x_2) are variables. The length of the column L is 250 cm. Material properties are: yield stress $\sigma_y = 500$ kgf/cm², modulus of elasticity $E = 0.85 \times 10^6$ kgf/cm², and density $\rho = 0.0025$ kgf/cm³.

Variables: $x = (d, t)$

Objective Function:

$$f(x) = 9.82dt + 2d$$

Constraints:

$$\begin{aligned}
g_1(x) &= \frac{P}{\pi d t \sigma_y} - 1 \leq 0 \\
g_2(x) &= \frac{8PL^2}{\pi^3 E d t (d^2 + t^2)} - 1 \leq 0 \\
g_3(x) &= \frac{2.0}{d} - 1 \leq 0 \\
g_4(x) &= \frac{d}{14} - 1 \leq 0 \\
g_5(x) &= \frac{0.2}{t} - 1 \leq 0 \\
g_6(x) &= \frac{t}{0.8} - 1 \leq 0
\end{aligned}$$

Variable Ranges:

$$2 \leq d \leq 14, \quad 0.2 \leq t \leq 0.8$$

VII. I-Beam Design Problem

The objective is to find the optimal dimensions to minimize the vertical deflection of an I-beam under predefined loads while satisfying cross-sectional area and stress constraints.

Variables: $x = (h, b, t_w, t_f) = (x_1, x_2, x_3, x_4)$

Objective Function:

$$f(x) = \frac{5000}{\frac{t_w(h-2t_f)^3}{12} + \frac{bt_f^3}{6} + 2bt_f \left(\frac{h-t_f}{2}\right)^2}$$

Constraints:

$$\begin{aligned}
g_1(x) &= 2bt_w + t_w(h - 2t_f) - 300 \leq 0 \\
g_2(x) &= \frac{18h \times 10^4}{t_w(h - 2t_f)^3 + 2bt_w(4t_f^2 + 3h(h - 2t_f))} - 1 \leq 0
\end{aligned}$$

Variable Ranges:

$$10 \leq h \leq 80, \quad 10 \leq b \leq 50, \quad 0.9 \leq t_w \leq 5, \quad 0.9 \leq t_f \leq 5$$

VIII. Three-Bar Truss Design Problem

The objective is to minimize the volume of a 3-bar truss subject to stress constraints on each side of the truss members.

Variables: $x = (A_1, A_2)$

Objective Function:

$$f(x) = (2\sqrt{2}A_1 + A_2) \times l$$

Constraints:

$$\begin{aligned}
g_1(x) &= \frac{\sqrt{2}A_1 + A_2}{\sqrt{2}A_1^2 + 2A_1A_2}P - \sigma \leq 0 \\
g_2(x) &= \frac{A_2}{\sqrt{2}A_1^2 + 2A_1A_2}P - \sigma \leq 0 \\
g_3(x) &= \frac{1}{A_1 + \sqrt{2}A_2}P - \sigma \leq 0
\end{aligned}$$

Variable Ranges:

$$0 \leq A_1, A_2 \leq 1$$

IX. Cantilever Beam Design Problem

This structural engineering problem involves weight optimization of a cantilever beam with a square cross-section. The beam is rigidly supported at one end, and a vertical force acts on the free node. The beam is composed of 5 hollow square blocks of constant thickness (2/3), while their height (or width) are the decision variables.

Objective Function:

$$f(X) = 0.0624(x_1 + x_2 + x_3 + x_4 + x_5)$$

Constraints:

$$g(X) = \frac{61}{x_1^3} + \frac{37}{x_2^3} + \frac{19}{x_3^3} + \frac{7}{x_4^3} + \frac{1}{x_5^3} - 1 \leq 0$$

Variable Ranges:

$$0.01 \leq x_i \leq 100, \quad i = 1, 2, \dots, 5$$

X. Piston Lever Design Problem

The main objective is to position the piston lever components $H(=x_1)$, $B(=x_2)$, $D(=x_3)$, and $X(=x_4)$ to minimize the oil volume when raising the lever from 0° to 45° .

Objective Function:

$$f(X) = \frac{1}{4}\pi x_3^2(L_2 - L_1)$$

Constraints:

$$\begin{aligned} g_1(X) &= QL \cos \theta - R \times F \leq 0 \\ g_2(X) &= Q(L - x_4) - M_{\max} \leq 0 \\ g_3(X) &= 1.2(L_2 - L_1) - L_1 \leq 0 \\ g_4(X) &= \frac{x_3}{2} - x_2 \leq 0 \end{aligned}$$

Where:

$$\begin{aligned} R &= \frac{|-x_4(x_4 \sin \theta + x_1) + x_1(x_2 - x_4 \cos \theta)|}{\sqrt{(x_4 - x_2)^2 + x_1^2}} \\ F &= \frac{\pi P x_3^2}{4} \\ L_1 &= \sqrt{(x_4 - x_2)^2 + x_1^2} \\ L_2 &= \sqrt{(x_4 \sin \theta + x_1)^2 + (x_2 - x_4 \cos \theta)^2} \end{aligned}$$

Parameters: $\theta = 45^\circ$, $Q = 10000$ lbs, $L = 240$ in, $M_{\max} = 1.8 \times 10^6$ lbs·in, $P = 1500$ psi.

Variable Ranges:

$$0.05 \leq x_1, x_2, x_4 \leq 500, \quad 0.05 \leq x_3 \leq 200$$

XI. Corrugated Bulkhead Design Problem

This problem aims to minimize the weight of a chemical tanker's corrugated bulkhead. The design variables are width (x_1), depth (x_2), length (x_3), and plate thickness (x_4).

Objective Function:

$$f(X) = \frac{5.885x_4(x_1 + x_3)}{x_1 + \sqrt{x_3^2 - x_2^2}}$$

Constraints:

$$\begin{aligned}
g_1(X) &= -x_4x_2 \left(0.4x_1 + \frac{x_3}{6}\right) + 8.94 \left(x_1 + \sqrt{x_3^2 - x_2^2}\right) \leq 0 \\
g_2(X) &= -x_4x_2^2 \left(0.2x_1 + \frac{x_3}{12}\right) + 2.2 \left(8.94 \left(x_1 + \sqrt{x_3^2 - x_2^2}\right)\right)^{4/3} \leq 0 \\
g_3(X) &= -x_4 + 0.0156x_1 + 0.15 \leq 0 \\
g_4(X) &= -x_4 + 0.0156x_3 + 0.15 \leq 0 \\
g_5(X) &= -x_4 + 1.05 \leq 0 \\
g_6(X) &= -x_3 + x_2 \leq 0
\end{aligned}$$

Variable Ranges:

$$0 \leq x_1, x_2, x_3 \leq 100, \quad 0 \leq x_4 \leq 5$$

XIII. Reinforced Concrete Beam Design Problem

Assuming a reinforced concrete beam has a span of 30 ft and sustains a live load of 2000 lbf and a dead load of 1000 lbf (including the beam's weight). The compressive strength of concrete (σ_c) is 5 ksi, and the yield stress of steel (σ_y) is 50 ksi. The cost of concrete is \$0.02 / in² / linear ft, and the cost of steel is \$1.0 / in² / linear ft. The depth-to-width ratio is restricted to less than or equal to 4.

Variables: Area of reinforcement A_s (x_1), beam width b (x_2), and beam depth h (x_3).

Objective Function:

$$f(X) = 2.9x_1 + 0.6x_2x_3$$

Constraints:

$$\begin{aligned}
g_1(X) &= \frac{x_2}{x_3} - 4 \leq 0 \\
g_2(X) &= 180 + 7.375 \frac{x_1^2}{x_3} - x_1x_2 \leq 0
\end{aligned}$$

Variable Ranges:

$$0 \leq x_1, x_2 \leq 1, \quad 5 \leq x_3 \leq 10$$

Appendix: Penalty Function Method for Inequalities

This method is utilized to handle inequality constraints by incorporating the constraint violations as part of the objective function. The corresponding augmented penalty function is:

$$J_{\text{aug}}(x) = f(x) + \sum_{i=1}^n k_i \cdot b_i$$

Where k_i represents the penalty factor, and b_i is calculated as:

$$b_i = \begin{cases} 0 & \text{if } x_i \leq 0 \\ x_i^2 & \text{if } x_i > 0 \end{cases} \quad i = 1, 2, \dots, n$$

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